Seria 9 - Algaed - CD312 - 2.12.2021.

Ortodiagonalizarea matricilor simetrice.

(T) $A \in M_n(\mathbb{R})$; $A = A^t \Rightarrow \exists Q \in M_n(\mathbb{R})$; Q ortogonală ;
$\exists D$ mat diagonală a.î. $\boxed{A = QDQ^t}$.

Def: $Q \in M_{m,n}(\mathbb{R})$ s.n. **ortogonală** dc. $\underset{m\times n}{Q} \cdot \underset{n\times m}{Q^t} = I_m$; $Q^tQ = I_n$.

Prop. Q matr ortogonală ; $Q = [q_1 | .. | q_n] \in M_{m,n}(\mathbb{R}) \Leftrightarrow \{q_1, .., q_n\}$ **ortonormată**
$\|q_i\| = 1$; $\forall i$; $q_i \perp q_j$; $\forall i \neq j$.

Prop. $A = A^t$; $\lambda_1 \neq \lambda_2$ val. proprii $\Rightarrow V_1 \perp V_2$ (vectorii proprii)
(V_1 ← λ_1 ; V_2 ↓ λ_2)

Ex 1: $A = \begin{pmatrix} 10 & 20 \\ 20 & 40 \end{pmatrix}$ $A = A^t$. Orto-diagonalizați A.

(I) Diagonalizarea mat A. $A \in M_n(\mathbb{R})$ → ? val. proprii ; ? vectorii proprii.
$A = PDP^{-1}$; $D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}$; $P = [v_1 | .. | v_n]$

(II) $\{v_1, .., v_n\} \xrightarrow{\text{Gram-Schmidt}} \{q_1, .., q_n\}$. bază ortonormată.

$Q = [q_1 | .. | q_n] \Rightarrow \boxed{A = QDQ^t}$.

$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} 10-\lambda & 20 \\ 20 & 40-\lambda \end{vmatrix} = (10-\lambda)(40-\lambda) - 400 = \lambda^2 - 50\lambda$
$= \lambda(\lambda - 50)$.

$\begin{cases} \lambda_1 = 50 \\ \lambda_2 = 0 \end{cases} \Rightarrow \boxed{D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 50 & 0 \\ 0 & 0 \end{pmatrix}}$.

$\boxed{\lambda_1 = 50} \Rightarrow$? $v = \begin{pmatrix} x \\ y \end{pmatrix}$ a.î. $A \cdot v = \lambda_1 v \Rightarrow \begin{pmatrix} 10 & 20 \\ 20 & 40 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 50\begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow$

$\Rightarrow \begin{cases} 10x + 20y = 50x \\ 20x + 40y = 50y \end{cases} \Rightarrow 20y = 40x \Rightarrow \boxed{y = 2x} \Rightarrow \vec{v} = \begin{pmatrix} x \\ 2x \end{pmatrix} = x\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$\boxed{v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}} \Rightarrow V_{\lambda_1} = Sp\{v_1\}$; $\dim V_{\lambda_1} = 1$

$\boxed{\lambda_2 = 0} \Rightarrow A\vec{v} = \lambda_2 \cdot \vec{v} \Rightarrow A\vec{v} = \vec{0} \Rightarrow \begin{pmatrix} 10 & 20 \\ 20 & 40 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 10x + 20y = 0$
$\Rightarrow x = -2y$.

? $\vec{v}$ a.î.

$\Rightarrow \vec{v} = \begin{pmatrix} -2y \\ y \end{pmatrix} = y\begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow \boxed{\vec{v_2} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}}$.

$A = PDP^{-1}$
$P = [v_1 | v_2]$

$\{v_1, v_2\} \xrightarrow{G\text{-}S} \{q_1, q_2\}$ bază ortonormată pt $\mathbb{R}^2$.

bază pt $\mathbb{R}^2$

$\langle v_1, v_2 \rangle = 0$.

$\lambda_1 + \lambda_2 \Rightarrow v_1 \perp v_2$.

$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$; $v_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.; $\|v_1\| = \sqrt{5}$

$\|v_2\| = \sqrt{5}$

$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}$; $q_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$.

$\boxed{A = QDQ^t}$; $Q = (q_1 | q_2) = \begin{pmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{pmatrix}$.; $QQ^t = I_2 = Q^tQ$

2x2 2x2.

$A^n = QD^nQ^t$.

Ex2 ? complem. ortogonal pt $V_0 = Sp\{\overbrace{(1,1,0)}^{v_1}, \overbrace{(1,1,-1)}^{v_2}\} \subseteq \mathbb{R}^3$

Scrieți vect $v = (0,-2,1)$ ca o sumă de doi vectori, unul din V_0 și celălalt din $V_0^\perp$.

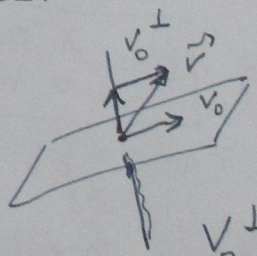
Met I · ? v_3 s.t. $v_3 \perp v_1$, $v_3 \perp v_2$ $\Rightarrow v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{cases} x+y=0 \\ x+y-z=0 \end{cases} \Rightarrow z=0$. $x=-y \Rightarrow v_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$\{v_1, v_2, v_3\}$ bază pt $\mathbb{R}^3$; $v \in \mathbb{R}^3 \Rightarrow$

$v = \underbrace{\alpha v_1 + \beta v_2}_{\in V_0} + \underbrace{\gamma v_3}_{\in V_0^\perp} \Rightarrow \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} = \alpha\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \gamma\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$.

$\Rightarrow \begin{cases} 0 = \alpha + \beta + \gamma \\ \dots \end{cases} \Rightarrow$ sistem de 3 ec $\rightarrow \alpha_1, \alpha_2, \alpha_3$ unic.

Met II. V_0 = plan; $V_0^\perp$ dr.



$v = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$.

$V_0^\perp = Sp\{v_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}\}$.

$\vec{v} = \underbrace{proj_{V_0^\perp}\vec{v}}_{\in V_0^\perp} + \underbrace{(\vec{v} - proj_{V_0^\perp}\vec{v})}_{\in V_0}$.

$proj_{V_0^\perp}\vec{v} = \frac{\langle \vec{v}, \vec{v_3} \rangle}{\langle \vec{v_3}, \vec{v_3} \rangle} \cdot v_3 = proj_{v_3}\vec{v} = \frac{2}{2}(1,-1,0)$. $\rightarrow \vec{v}$

$proj_{V_0^\perp}\vec{v} = \vec{v_3} = (1,-1,0) \Rightarrow \vec{v} - proj_{V_0^\perp}\vec{v} = (0,-2,1) - (1,-1,0) =$

$= (-1,-1,1)$.

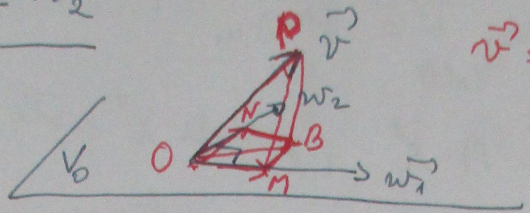
$\vec{v} = \underbrace{\vec{v_3}}_{\in V_0^\perp} + \underbrace{(\vec{v} - \vec{v_3})}_{\in V_0}$.

et III

$\{v_1, v_2\} \xrightarrow{G-S} \{w_1, w_2\}$ bază ortogonală.

$Sp\{v_1, v_2\} = Sp\{w_1, w_2\} = V_0$.

$w_1 \perp w_2$



$\vec{v} = \vec{OB} + \vec{BP} \in V_0^{\perp}$.

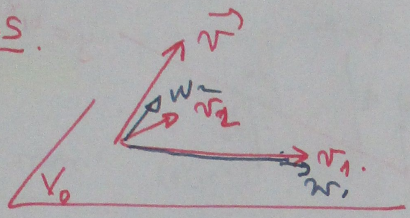
$\vec{OB} = \vec{OM} + \vec{ON}$.

$\boxed{\vec{OB} = proj_{w_1}\vec{v} + proj_{w_2}\vec{v}}$

$\in V_0$

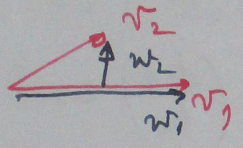
$\boxed{proj_{V_0}\vec{v} = proj_{w_1}\vec{v} + proj_{w_2}\vec{v}}$

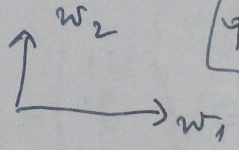
Obs.



$proj_{V_0}\vec{v} \neq proj_{v_1}\vec{v} + proj_{v_2}\vec{v}$.

$\{v_1, v_2\} \longrightarrow \{w_1, w_2\}$ ortogonali.

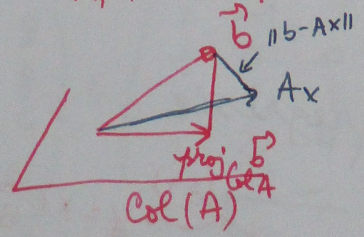




$\boxed{proj_{V_0}\vec{v} = proj_{w_1}\vec{v} + proj_{w_2}\vec{v}}$

Aproximarea soluției unui sist liniar incompatibil

$A \in \mathcal{M}_{m,n}(\mathbb{R})$; $b \in \mathbb{R}^m$; $b \notin Col(A)$.



$\underset{m\times n}{A} \cdot \underset{n\times 1}{\vec{x}} = \underset{m\times 1}{\vec{b}}$ incompatibil.

? $x \in \mathbb{R}^n$ î. $\|b - Ax\|$ min.

$\min \|b - Ax\| = \|b - proj_{Col A}\vec{b}\|$.

x

? $\vec{x}$ sol a sist compatibil. $\boxed{Ax = proj_{Col A}\vec{b}}$ $\Rightarrow$ x^*

sist $Ax = b$

$x^* = \begin{pmatrix} x_1^* \\ \vdots \\ x_n^* \end{pmatrix}$ pseudosol în sensul celor mai mici pătrate.

$\min \|b - Ax\| = \min \sqrt{\sum_{i=1}^{m} (b_i - \sum_{j=1}^{n} a_{ij} x_j)^2}$

? $proj_{Col A}\vec{b}$ în rap cu pr. scalar canonic.

? $\text{proj}_{\text{Col}A}\vec{b}$.

Met I) ? $\{u_1, .., u_r\}$ bază ortogonală pt $\text{Col}(A)$.

$\text{proj}_{\text{Col}A}\vec{b} = \text{proj}_{u_1} b + .. + \text{proj}_{u_r} b$

Met II) $\text{proj}_{\text{Col}A} b = A \cdot x^*$; unde x^* este sol pt.

$Ax = b$ incompat.

$\boxed{A^tAx = A^tb}$ sist. compatibil.

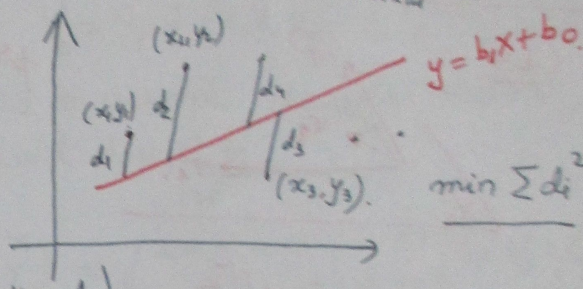
$x^* = (A^tA)^{-1}A^tb$.

$A^{n\times m}$ $m\times n$

Dreapta de regresie

(x_1, y_1) ; .. , (x_n, y_n).

? $\begin{pmatrix} b_1 \\ b_0 \end{pmatrix}$ i. $y = b_1x + b_0$.



$\min \sum d_i^2$.

$$\begin{cases} y_1 = b_1x_1 + b_0 \cdot 1 \\ y_2 = b_2x_2 + b_0 \cdot 1 \\ \vdots \\ y_n = b_nx_n + b_0 \cdot 1 \end{cases} \iff \underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} b_1 \\ b_0 \end{pmatrix}}_{\vec{x}} \to A\vec{x} = \vec{b}$$

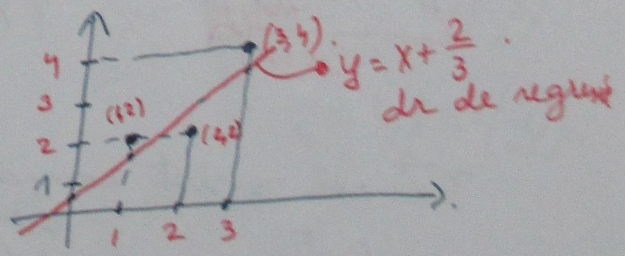
incompat.

$A^tA\vec{x} = A^tb$.

sist comp.

$\boxed{\hat{x} = (A^tA)^{-1}A^tb}$

Ex 3. $(1,2)$; $(2,2)$, $(3,4)$. ? dr de regresie.



$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_0 \end{pmatrix}.$$

$$\underbrace{\begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} b_1 \\ b_0 \end{pmatrix}}_{\vec{x}}.$$

$\vec{x} = (A^tA)^{-1}A^tb$. ; $A^tA = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}$

$y = b_1x + b_0$ dr de regresie.

$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow B^{-1} = \frac{1}{\det B} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \Rightarrow (A^tA)^{-1} = \frac{1}{42-36} \cdot \begin{pmatrix} 3 & -6 \\ -6 & 14 \end{pmatrix} \Rightarrow$

$(A^tA)^{-1} = \begin{pmatrix} 1/2 & -1 \\ -1 & 7/3 \end{pmatrix} \Rightarrow \vec{x} = (A^tA)^{-1} A^t \vec{b}$

$\vec{x} = \begin{pmatrix} 1/2 & -1 \\ -1 & 7/3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2/3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_0 \end{pmatrix} \Rightarrow \boxed{\begin{matrix} b_1 = 1 \\ b_0 = 2/3 \end{matrix}}$

$\boxed{y = x + \frac{2}{3}}$ dr de regresie.

↳ $(1, 1+\frac{2}{3})$; $(2, 2+\frac{2}{3})$; $(3, 3+\frac{2}{3})$.

Ex 4 ? parabolă de regresie.

$y = \alpha + \beta x + \gamma x^2$.

$(-1, 8)$; $(0, 8)$; $(1, 4)$, $(2, 16)$.

$\begin{cases} y_1 = \alpha \cdot 1 + \beta \cdot x_1 + \gamma \cdot x_1^2 \\ y_2 = \alpha \cdot 1 + \beta \cdot x_2 + \gamma \cdot x_2^2 \\ y_3 = \dots \\ y_4 = \alpha \cdot 1 + \beta \cdot x_4 + \gamma \cdot x_4^2 \end{cases}$

? α, β, γ.

$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}_{\vec{x}}$

$\underbrace{\begin{pmatrix} 8 \\ 8 \\ 4 \\ 16 \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}_{\vec{x}}$.

$A\vec{x} = \vec{b}$.

$A^tA\vec{x} = A^t\vec{b}$.

$\boxed{\vec{x} = (A^tA)^{-1} \cdot A^t\vec{b} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}$.

Factorizare Q·R.

$A = Q \cdot R$ ⇒ utilă pt a găsi pe cale val. proprii ale unei mat.
a rez sist lin.
a da o aproximare prin met alor mai mici pătr.

T: $A: m \times n$

$A = [v_1 | \dots | v_n]$; $\{v_1, \dots, v_n\}$ lin. ind.

$\Rightarrow A = Q \cdot R$ unde $Q: m \times n$; $Q = [w_1 | \dots | w_n]$; $\{w_1, \dots, w_n\}$ bază ortonormată pt $Col(A)$.

$A_{m \times n} = Q_{m \times n} \cdot R_{n \times n}$

R = $n \times n$ matr nesing superior triunghiulară)

$R = (r_{ij})$; $r_{ij} = \langle w_i, v_j \rangle$

Ex 4 $A = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \Rightarrow A = QR$?

$v_1 \quad v_2 \quad v_3$.

$Q: 4 \times 3$.

$R: 3 \times 3$.

$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix}$.

$$R = \begin{pmatrix} \langle v_1, w_1 \rangle & \langle w_1, v_2 \rangle & \langle w_1, v_3 \rangle \\ 0 & \langle w_2, v_2 \rangle & \langle w_2, v_3 \rangle \\ 0 & 0 & \langle w_3, v_3 \rangle \end{pmatrix}.$$

Desc. QR. pt rez sist. incompatibile.

$Ax = b \Rightarrow QR \cdot x = b$.

$A = QR \Rightarrow A^t = R^t Q^t$.

$(QR)^t QRx = (QR)^t b$.

$R^t Q^t Q R x = R^t Q^t b$.

$R^t R x = R^t Q^t b$. ; $\det R^t \neq 0$.

$\boxed{Rx = Q^t b} \Rightarrow \hat{x}$.

R superior triungh. ⇒ sist se rez prin înlocuire pas cu pas backward.

Ex 5 $QR = ?$ $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$; $v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$.

$\{v_1, v_2\} \xrightarrow{G-S} \{w_1, w_2\}$.

$w_1 = v_1 \cdot \frac{1}{\|v_1\|} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$.

$u_2 = v_2 - \text{proj}_{u_1} v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \frac{2-3}{2} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$; $R = \begin{pmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{pmatrix}$.